

Hilda Adwubi Osei

hildaosei109@gmail.com | website | google scholar | linkedin | github

Research Interests

- Themes: Robustness under distributional shift, multi-agent systems and information asymmetry, fairness in algorithmic and resource allocation.
- Methods: Convex optimization, agent-based and game-theoretic simulation, statistical and predictive modeling (regression, ensemble methods, transfer learning).
- Applications: Agricultural markets and economics, public health (infection prevention, environmental health).

Education

Kwame Nkrumah University of Science and Technology
B.Sc. Industrial Engineering

Kumasi, Ghana
2021 – 2024

Publications

Work in Progress

WP1. Osei, H.A. *FarmDirect: Data-Driven pricing and risk-adjusted allocation for Ghanaian tomato*. Manuscript submitted to arXiv (pending moderation review, cross-disciplinary classification); targeting journal submission.

Conference and Poster Presentations

CP1. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University*. Poster presentation, Korle Bu Teaching Hospital, Ghana, 2026.

Preprints and Technical Reports

PT2. Osei, H.A., Osei, C.T., Asobayire, D.Y. *Learning from waste: Machine Learning for health risk prediction and computer vision-based sorting in Ghana*. arXiv:2608.25759 [cs.LG] (cross-listed cs.CV, cs.CY), 2026.

PT1. Osei, C.T., Asare, A.O., Agyen, Y.O., Osei, H.A., Amooba, P.A. *Knowledge and Compliance with Standard Precautions for Nosocomial Infection Prevention among Undergraduate Nursing and Midwifery Students at a Ghanaian University*. medRxiv, 2026.

Research and Work Experience

FarmDirect

Founder & Lead Developer

Accra, Ghana
2025 – Present

- Designing a data-driven marketplace to connect farmers directly to buyers, eliminating middlemen and optimizing prices; conducted the underlying research through predictive modeling, risk-adjusted optimization, and agent-based simulation on real government market data.
- Selected as 1 of 9 finalists to pitch before a national judging panel, winning an Innovation Grant (GHS 10,000); presented at the “We, The Youth” Global Youth Summit (South Korea, 1 of 150 selected from 27,000+ applicants).

Ghana Atomic Energy Commission (GAEC)

Research & Innovation Collaborator

Accra, Ghana
Jun 2025 – Present

- Collaborate with GAEC on agricultural research applying AI and optimization to real-world farming challenges.

- Independently designed and led a full research pipeline (regression, clustering, risk-adjusted optimization, and agent-based simulation) analyzing agricultural commodity pricing, culminating in a self-authored technical report and preprint.

Headstarter AI

Software Engineering Fellow

Remote

Jul – Sep 2024

- Built and deployed 5 AI-powered applications in 5 weeks, including a RAG-based support agent (OpenAI + Pinecone) and an LLM-driven SaaS product (Llama 3.1 via Groq).
- Applied agile, CI/CD, and product-scaling practices under mentorship from engineers at Google, Y Combinator, and Stanford.

Teaching Experience

Independent Technology Education Initiative

Programming Mentor

Ghana

2026

- Design and deliver project-based programming curriculum for early-career learners, emphasizing applied problem-solving over rote instruction.
- Guide mentees through independent coding projects, building their debugging skills and technical confidence.

National Software Development Summer School, CTVET/GAEC

Teaching Assistant

Ghana

2025

- Deliver peer-led tutorials on full-stack development (Django, Flask) to a national cohort of aspiring developers.
- Mentor participants on project architecture, version control, and pitch preparation for national innovation competitions.

MATH 152: Calculus with Analysis, Kwame Nkrumah University of Science and Technology

Teaching Assistant

2024

- Led recitation sessions translating abstract analytical concepts into problem-solving strategies for undergraduates.
- Provided individualized feedback through grading and office hours to strengthen students' mathematical reasoning.

ME 259: Applications of Computer Graphics, Kwame Nkrumah University of Science and Technology

Teaching Assistant

Kumasi, Ghana

2022

- Supported students in applying computational graphics principles to engineering design problems through lab instruction.
- Reviewed and gave technical feedback on coding assignments to reinforce correct implementation practices.

Honors and Awards

- IFWY Global Explorer, International “We, The Youth” Forum, South Korea, 2025.
- Innovation & Sustainability Finalist, Mastercard Foundation x GAEC Challenge, 2025.
- GAEC & Mastercard Foundation Research Collaboration Recognition, 2025.
- STEM Women Ghana Changemaker Award, 2024.
- Academic Distinction, Second Class Upper, KNUST, 2024.
- TechWomen Africa Rising Recognition, 2023.
- Youth Arise Leadership Fellowship & Counselor Appointment, 2020.

Invited Talks

- Economic Transformation and Youth Inclusion. International “We, The Youth” Forum, UNRISD, Botswana, 2025.
- Economic Transformation and Youth Inclusion. International “We, The Youth” Forum, UNRISD, South Korea, 2025.
- Youth Engagement in Climate-Induced Loss and Damage in Agrifood Systems. World Food Forum, 2025.